Fundamentals Standards And Codes Types Materials Applications Resources Capacitor

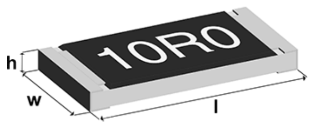
Resistor standards and codes » Resistor Sizes and Packages

Resistor Sizes and Packages

Resistors are available in a large amount of different packages. Nowadays the most used are the rectangular surface mount resistors, but also the good old axial resistor is still used extensively in through hole designs. This page will inform you about the dimensions of SMD, axial and MELF packages and about the required land patterns for SMD components.

SMD resistor sizes

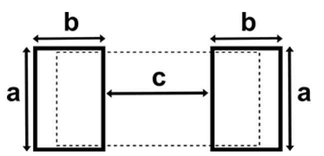
The shape and size of surface mount resistors are standardized, most manufacturers use the JEDEC standards. The size of SMD resistors is indicated by a numerical code, such as 0603. This code contains the width and height of the package. So in the example of 0603 Imperial code, this indicates a length of 0.060″ and a width of 0.030″. This code can be given in Imperial or Metric units, in general the Imperial code is used more often to indicate the package size. On the contrary in modern PCB design metric units (mm) are more often used, this can be confusing. In general you can assume the code is in imperial units, but the dimensions used are in mm. The SMD resistor size depends mainly on the required power rating. The following table lists the dimensions and specifications of commonly used surface mount packages.



Code		Length (l)		Width (w)		Height (h)		Power
Imperial	Metric	inch	mm	inch	mm	inch	mm	Watt
0201	0603	0.024	0.6	0.012	0.3	0.01	0.25	1/20 (0.05)
0402	1005	0.04	1.0	0.02	0.5	0.014	0.35	1/16 (0.062)
0603	1608	0.06	1.55	0.03	0.85	0.018	0.45	1/10 (0.10)
0805	2012	0.08	2.0	0.05	1.2	0.018	0.45	1/8 (0.125)
1206	3216	0.12	3.2	0.06	1.6	0.022	0.55	1/4 (0.25)
1210	3225	0.12	3.2	0.10	2.5	0.022	0.55	1/2 (0.50)
1218	3246	0.12	3.2	0.18	4.6	0.022	0.55	1
2010	5025	0.20	5.0	0.10	2.5	0.024	0.6	3/4 (0.75)
2512	6332	0.25	6.3	0.12	3.2	0.024	0.6	1

Solder pad land pattern

When designing with surface mount components the right solder pad size and land pattern should be used. The following table shows an indication of the land pattern for common surface mount packages. The table lists the dimensions for reflow soldering, for wave soldering smaller pads are used.



Code		Pad length (a)		Pad width (b)		Gap (c)	
Imperial	Metric	inch	mm	inch	mm	inch	mm
0201	0603	0.012	0.3	0.012	0.3	0.012	0.3
0402	1005	0.024	0.6	0.020	0.5	0.020	0.5
0603	1608	0.035	0.9	0.024	0.6	0.035	0.9
0805	2012	0.051	1.3	0.028	0.7	0.047	1.2

Contents

Popular pages

- What is a resistor?
- Resistor color code
  - Color code calculator
- Varistor
- NTC thermistor
- Potentiometer

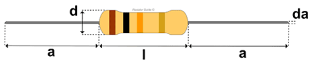
Tags

0201 resistor, 0402 resistor, 0603 resistor, 0805 resistor, 1005 resistor, 1206 resistor, 1210 resistor, 1608 resistor, 2012 resistor, 2512 resistor, 3216 resistor, 3246 resistor, 6332 resistor, axial resistor, JEDEC, MELF, MELF resistor, microMELF resistor, miniMELF resistor, packages, resistor, resistor dimensions, resistor size, sizes, SMD resistor, solder pad pattern

1206	3216	0.063	1.6	0.035	0.9	0.079	2.0
1218	3246	0.19	4.8	0.035	0.9	0.079	2.0
2010	5025	0.11	2.8	0.059	0.9	0.15	3.8
2512	6332	0.14	3.5	0.063	1.6	0.15	3.8

Axial resistor size

The size of axial resistors is not as standardized as the SMD resistors and different manufacturers often use slightly different dimensions. Furthermore the size of an axial resistor depends on the power rating and the type of resistor such as carbon composition, wirewound, carbon or metal film. The following drawing and table give an indication of the dimensions of common carbon film and metal film axial resistors. Whenever the exact size needs to be known, always check the manufacturer datasheet of the component.



Power rating	Body length (l)	Body diameter (d)	Lead length (a)	Lead diameter (da)
Watt	mm	mm	mm	mm
1/8 (0.125)	3.0 ± 0.3	1.8 ± 0.3	28 ± 3	0.45 ± 0.05
1/4 (0.25)	6.5 ± 0.5	2.5 ± 0.3	28 ± 3	0.6 ± 0.05
1/2 (0.5)	8.5 ± 0.5	3.2 ± 0.3	28 ± 3	0.6 ± 0.05
1	11 ± 1	5 ± 0.5	28 ± 3	0.8 ± 0.05

MELF resistor package sizes

Sometimes surface mount resistors are also used as MELF packages (Metal Electrode Leadless Face). The main advantage of using MELF in stead of standard SMD packages is the lower thermal coefficient and better stability. The TCR of thin film MELF resistors is often between 25-50 ppm/K while standard thick film SMD resistors often have a TCR of > 200 ppm/K. This is possible due to the cylindrical construction of MELF resistors. This cylindrical construction also gives the package distinct disadvantages, mainly when the components have to be placed using pick and place machines. Because of their round shape a special suction cup and more vacuum is required. There are three common MELF package sizes: MicroMELF, MiniMELF and MELF. The following table lists the characteristics of these types.

Name	Abbr.	Code	Length	Diameter	Power
			mm	mm	Watt
MicroMELF	MMU	0102	2.2	1.1	0.2 - 0.3
MiniMELF	MMA	0204	3.6	1.4	0.25 - 0.4
MELF	MMB	0207	5.8	2.2	0.4 - 1.0

Resources

Online

- JEDEC : JEP95 - standardized component outlines